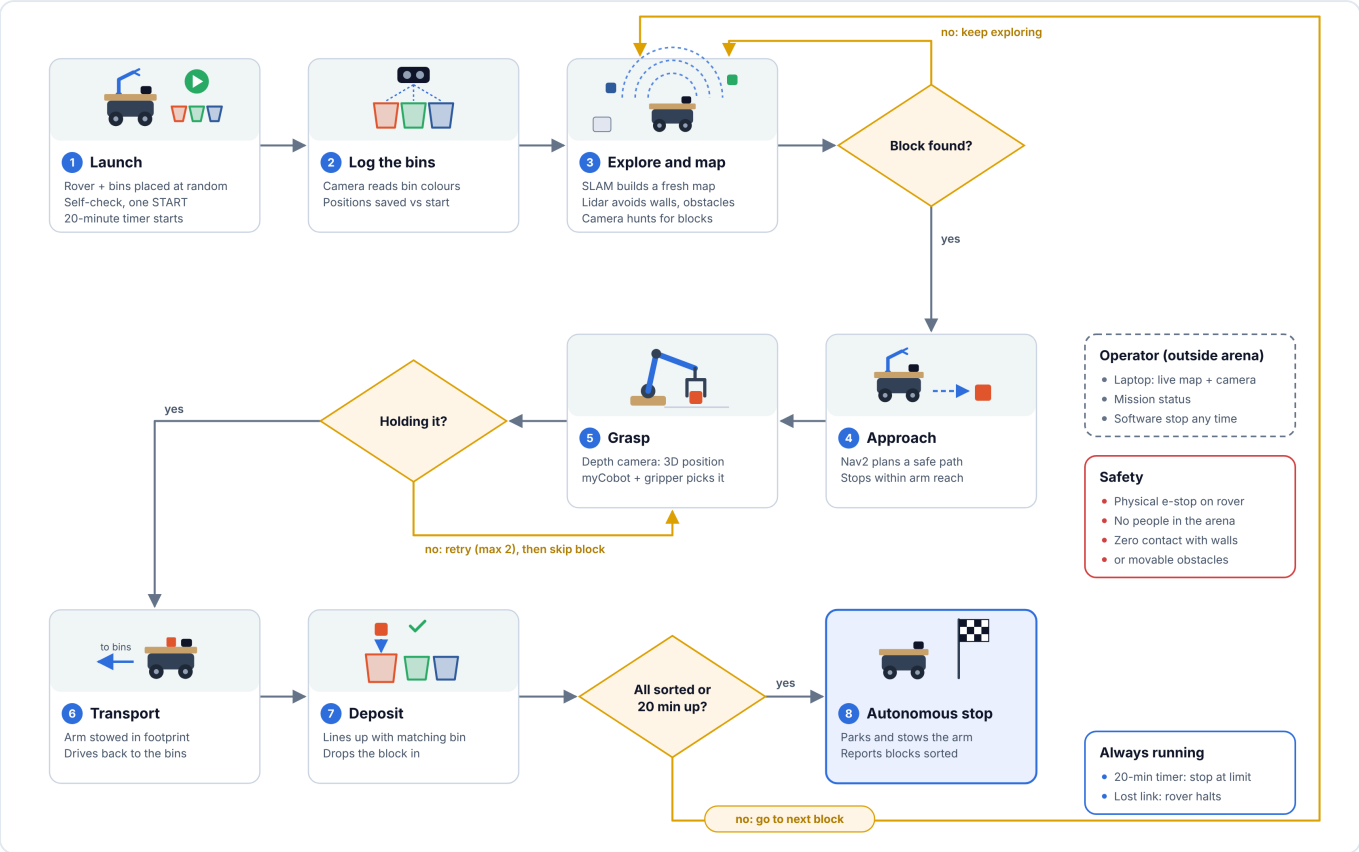
Concept of Operations

Autonomous Mobile Manipulation & Block Sorting System (LeoRover) · AERO62520 · CWK3 Design Requirements Analysis · Team 6 · November 2026

After one start command, the LeoRover autonomously finds, picks up and sorts 3 coloured blocks into matching bins within 20 minutes, in a randomised arena, with zero collisions.



Step-by-step overview

1 Launch

- Rover and bins placed at random; self-check, then one START.
- The 20-minute timer starts.

2 Log the bins

- Camera reads each bin's colour and saves its position.

3 Explore and map

- SLAM builds a fresh map; lidar avoids walls and obstacles.
- No block found: keep exploring.

4 Approach

- Nav2 plans a safe path and stops within arm reach.

5 Grasp

- Depth camera finds the block's 3D position; the arm picks it up.
- Not held: retry up to twice, then skip the block.

6 Transport

- Arm stowed; rover drives back to the bins, avoiding obstacles.

7 Deposit

- Lines up with the matching bin and drops the block in.
- Blocks left: go to the next block (step 3).

8 Autonomous stop

- All 3 sorted or 20 min up: parks, stows the arm, reports.

Throughout: operator watches from outside the arena with a software stop; physical e-stop on the rover; the rover halts if the laptop link is lost; custom sled in MDF, PLA/PETG, ABS or acrylic/PVC only.